

Interval notation



1 Write the interval notation for each of the following pictured inequalities:

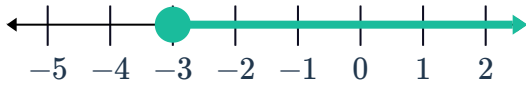
a



b



c



d



e



2 For each of the following inequalities express the solutions of the inequality using interval notation:

a $x < -7$

b $x \leq 6$

c $x \geq -4$

d $-7 \leq x \leq 3$



Domain and range

3 For each of the following relations:

- i Find the domain.
- ii Find the range.
- iii Determine whether the relation is a function or not.

a $\{(8, 5), (1, 3), (3, 4), (9, 1), (2, 9)\}$

b $\{(2, 3), (3, 8), (6, 1), (8, 7), (3, 2)\}$

c $\{(4, 4), (8, 9), (2, 8), (7, 9), (3, 1)\}$

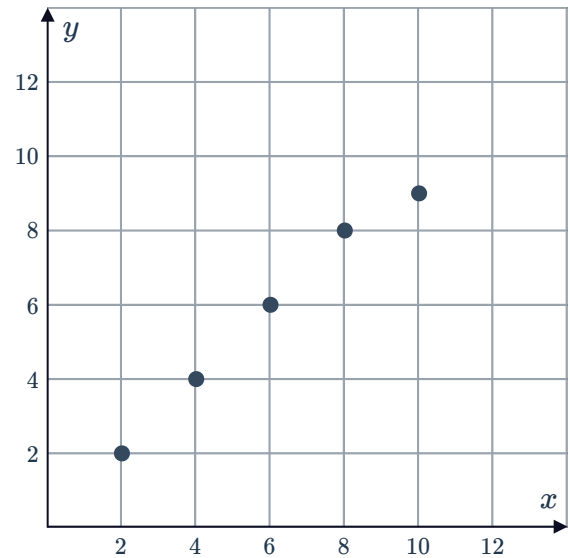
d

x	1	6	3	8	2
y	3	2	7	1	2

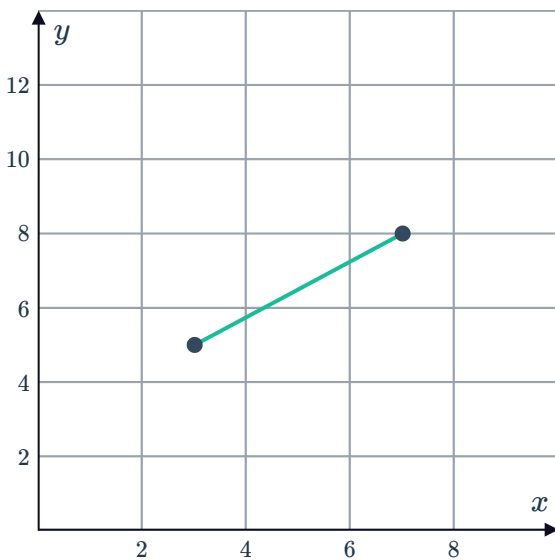
e

x	7	7	8	5	3
y	1	9	3	2	6

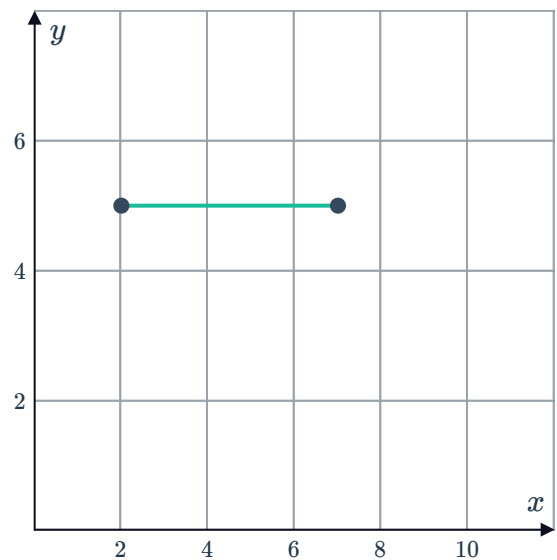
f



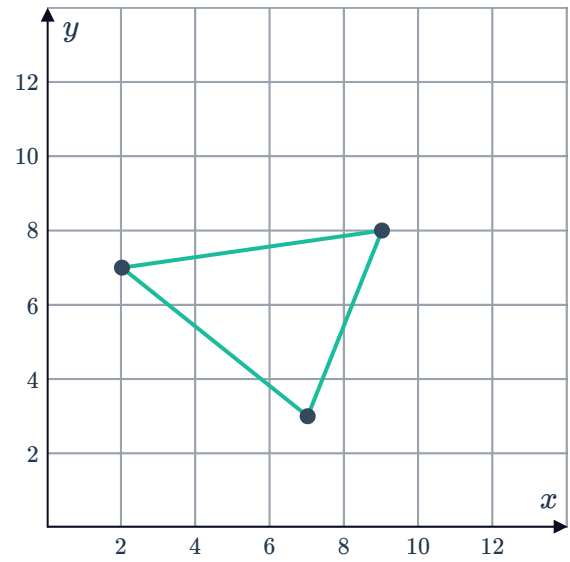
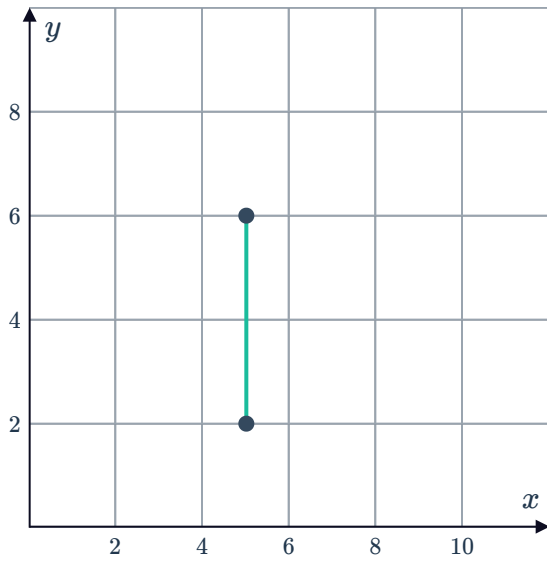
g



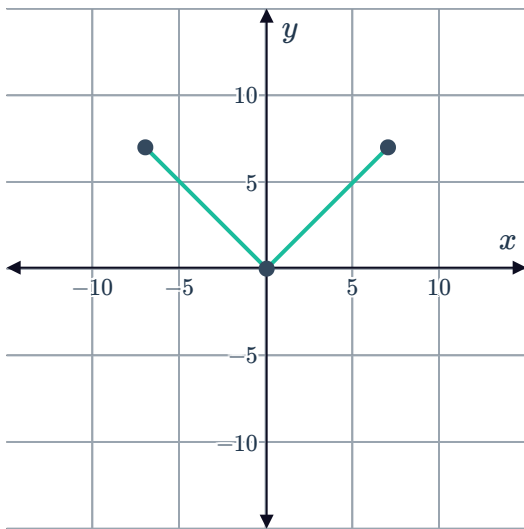
h



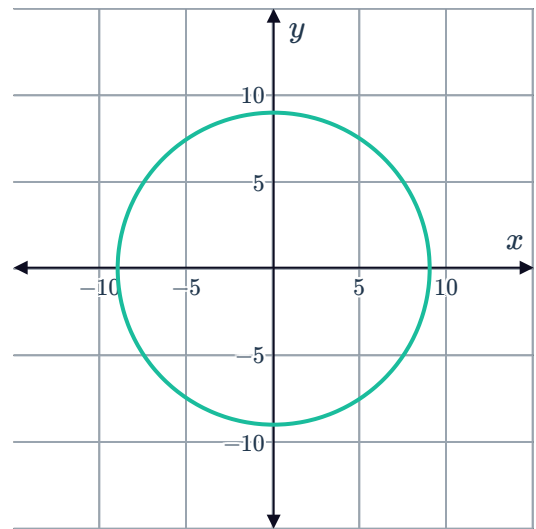
i



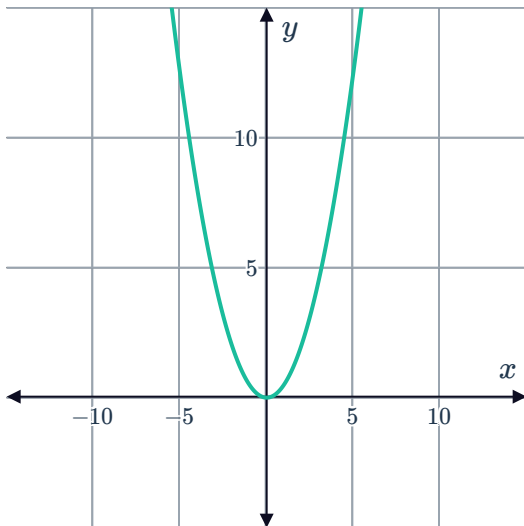
k



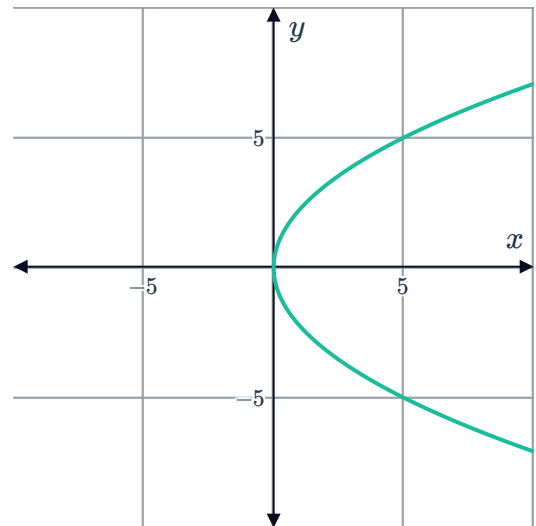
l



m



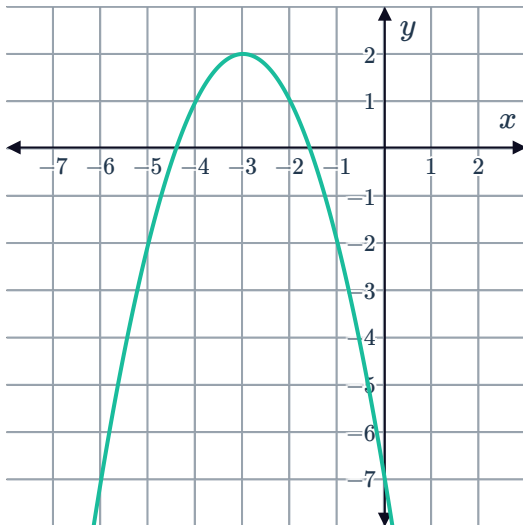
n



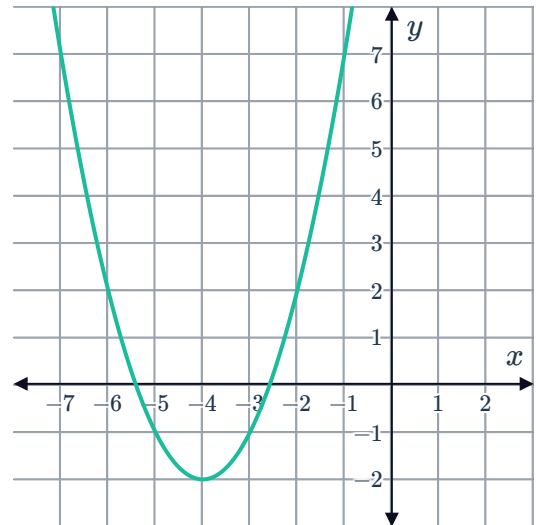
4 State the domain and range of the following functions:



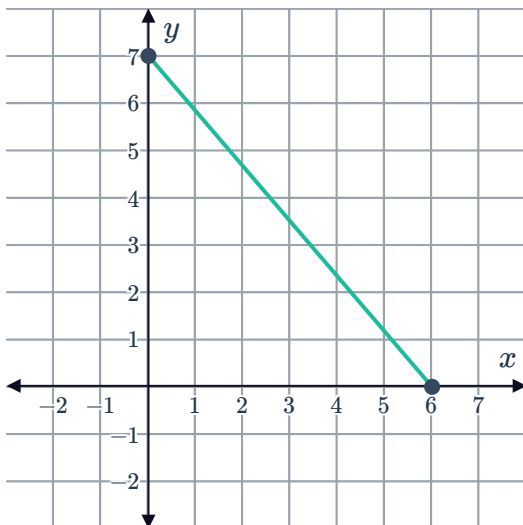
a



b



c



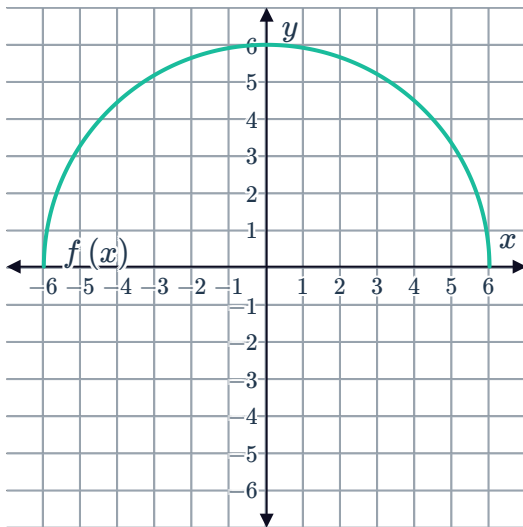
5 State the domain of the following functions



a $f(x) = -3x - 1$

b $f(x) = 8x + 8$

c



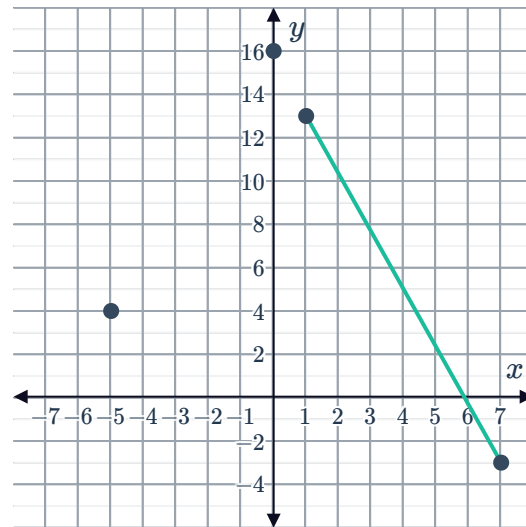
6 Consider the function f graphed. Use the graph to determine whether each of the given statements are true or false:

a The domain is $[-5, 7]$

b The range is $[-3, 16]$

c $f(1) - f(7) = 16$

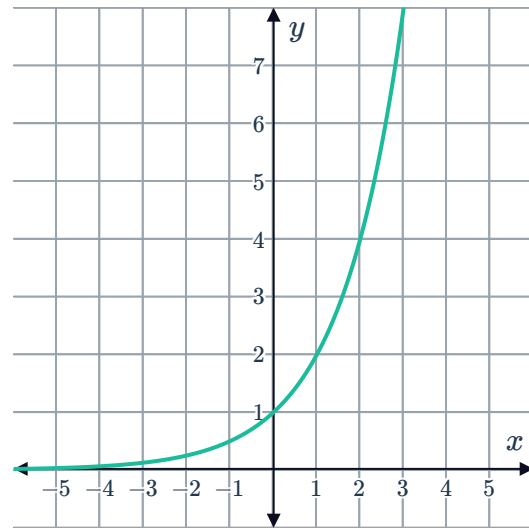
d $f(0) = 16$



7 Consider the graph of $y = 2^x$:

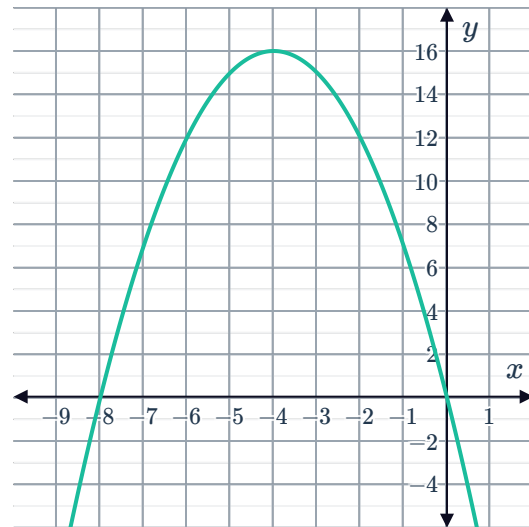


- a State the y -intercept of the graph.
- b Does the graph have an x -intercept?
- c State the domain of the function.
- d State the range of the function.
- e Find the value of y when $x = 7$.
- f Find the value of x when $y = 256$.



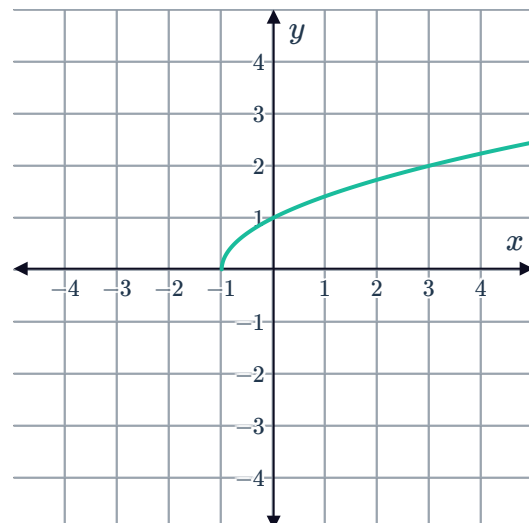
8 Consider the graph of the function shown:

- a State the maximum value of the graph.
- b State the range of the function.
- c State the domain of the function.



9 Consider the graph of the function $f(x) = \sqrt{x+1}$:

- a State the domain of the function.
- b Is there a value in the domain that can produce a function value of -2 ?



10 Consider the function $f(x) = (x - 2)^2$, for all x .



- a Graph the function.
- b Is there a value in the domain that can produce a function value of -9 ?

Continuity

11 A function $f(x)$ is known to be continuous on the interval $(-\infty, \infty)$.

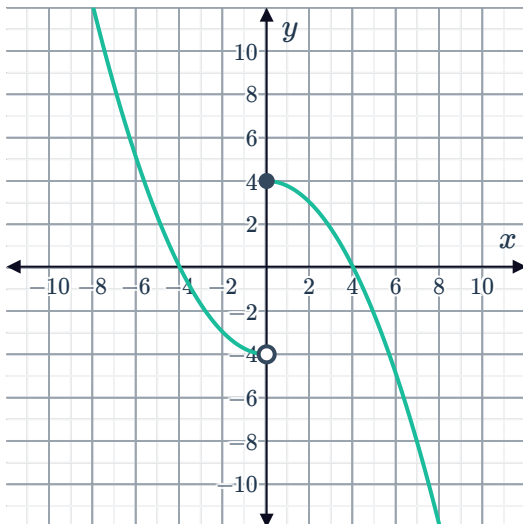
- a Is $f(x)$ continuous over all smaller intervals?
- b Suppose the function is $f(x) = 3x + 8$. State whether $f(x)$ is continuous on the following intervals:
 - i $[4, 6]$
 - ii $(-7, -1)$
 - iii $(0, \infty)$

12 Consider the function $f(x) = 4x^3 - 9x^2 - 2x + 1$. State whether $f(x)$ is continuous over the following intervals:

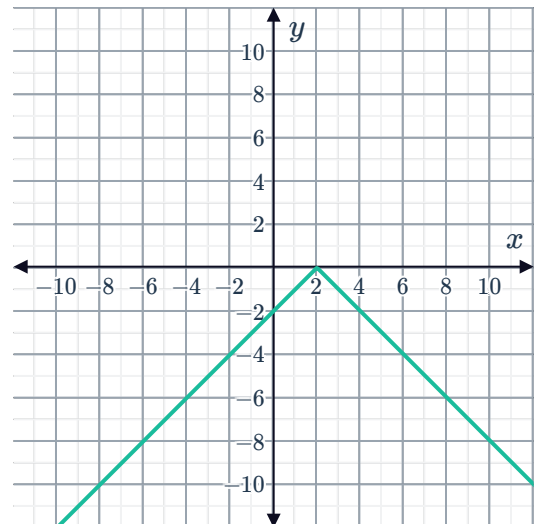
- a $(-\infty, \infty)$
- b $[9, 15]$

13 State whether the following functions are continuous:

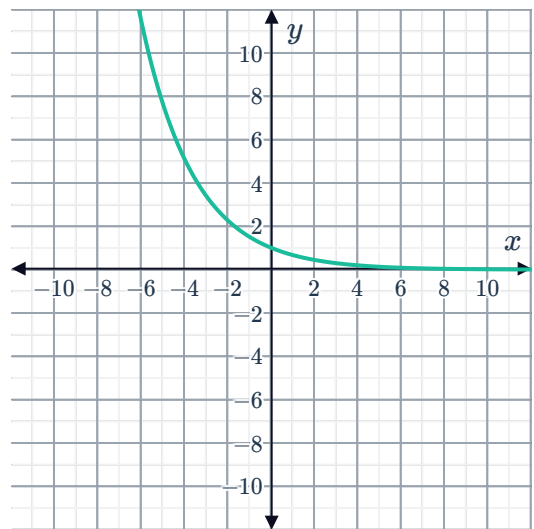
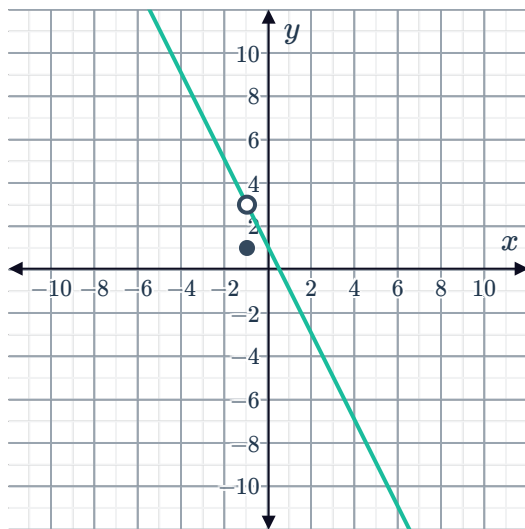
a



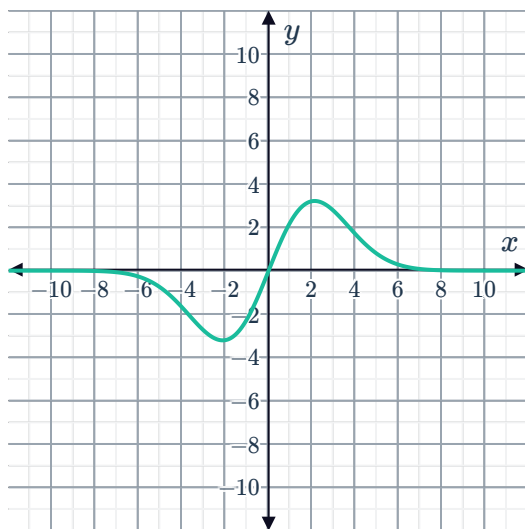
b



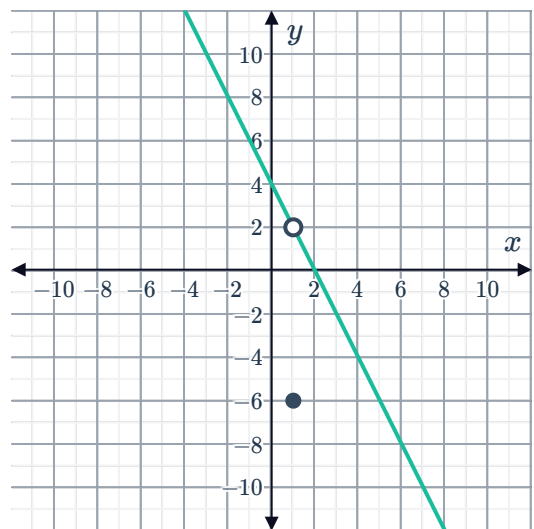
c



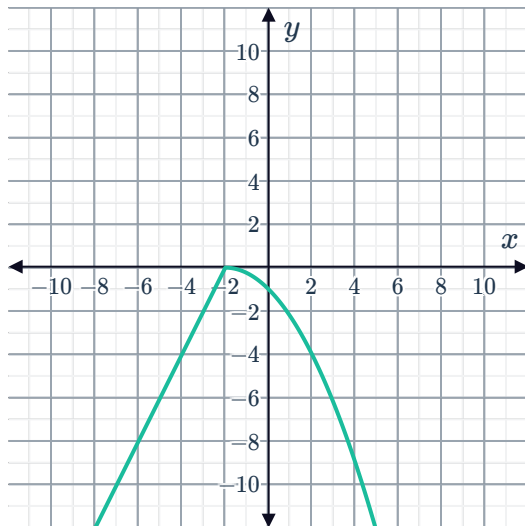
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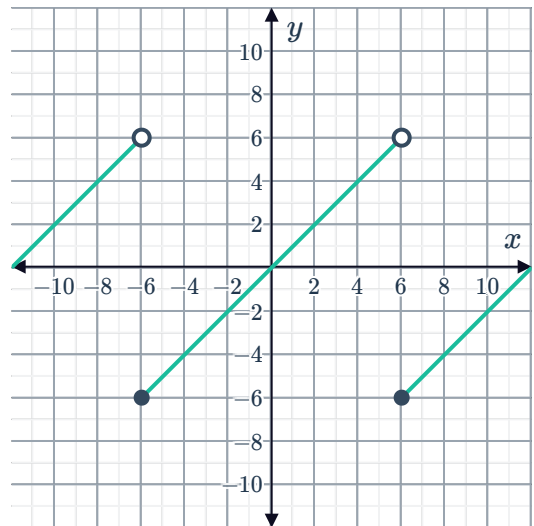
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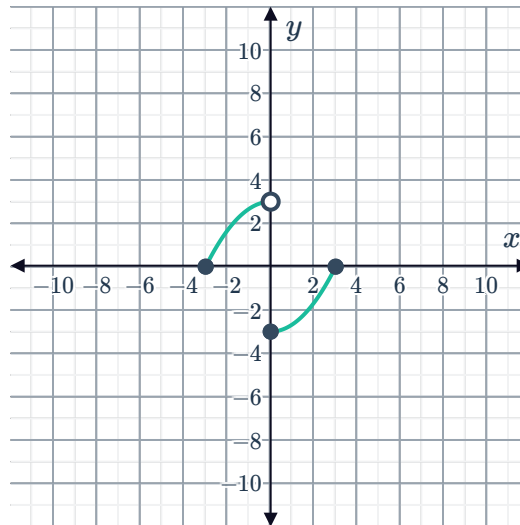
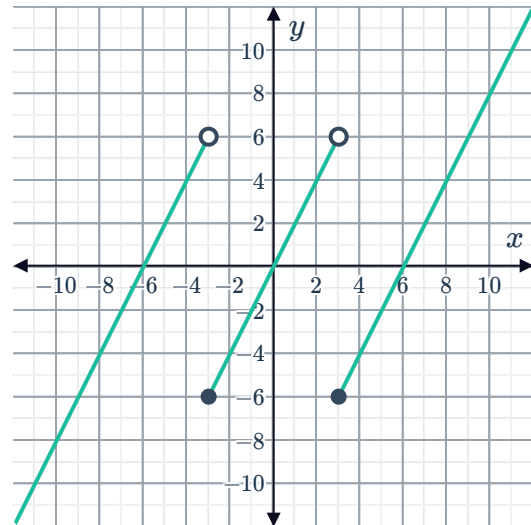
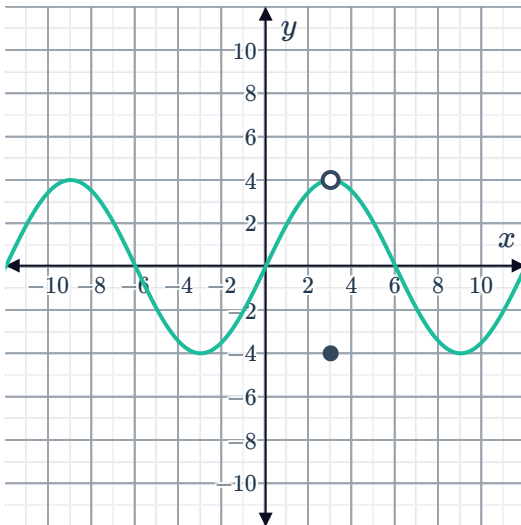
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h



14 Consider the following graphs of functions:

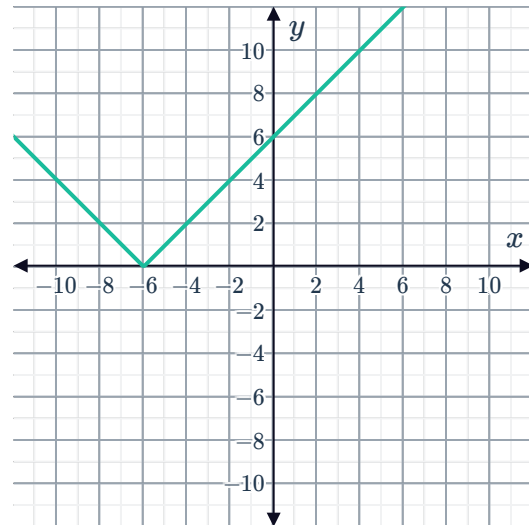


State whether the following statements are true or false:

- a Some of the functions are continuous.
- b All of the functions have at least one discontinuous point in their domain.
- c All of the functions have a domain of $(-\infty, \infty)$.

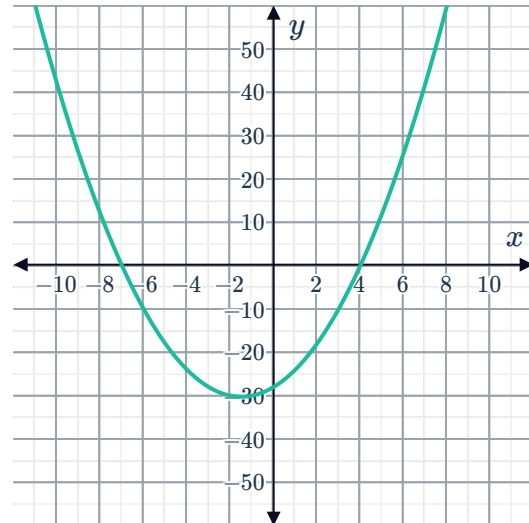
- 15 Consider the graph of the function $f(x) =$ :
State whether the function is continuous
over the following intervals:

- a $(-6, \infty)$ b $(-\infty, -6)$
c $(-\infty, \infty)$



- 16 Consider the graph of the function $f(x) = (x + 7)(x - 4)$:
State whether $f(x)$ is a continuous function
over:

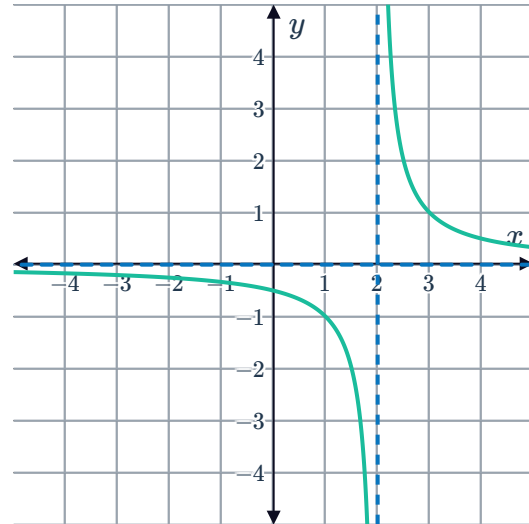
- a The domain.
b The interval $[-7, 4]$



17 Consider the graph of the function $f(x) = \frac{1}{x-2}$

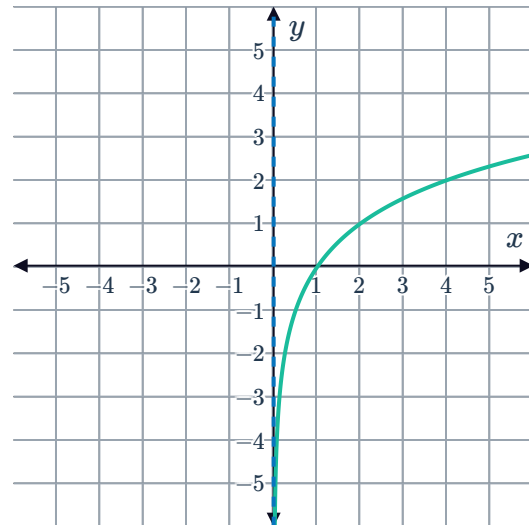


- a State the domain of $f(x)$.
- b Is $f(x)$ continuous over the interval $(-\infty, 2) \cup (2, \infty)$?
- c Is $f(x) = \frac{1}{x-2}$ a continuous function over its domain?



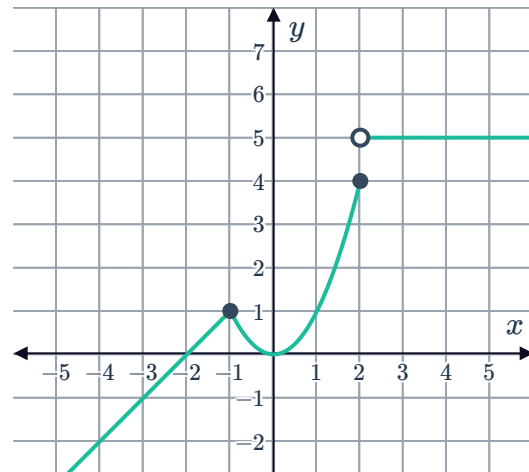
18 Consider the graph of the function $f(x) = \log_2 x$:

- a State the domain of $f(x)$.
- b Is $f(x)$ continuous over the interval $(0, \infty)$?
- c Is $f(x) = \log_2 x$ a continuous function over its domain?



19 Consider the function $f(x)$ shown:

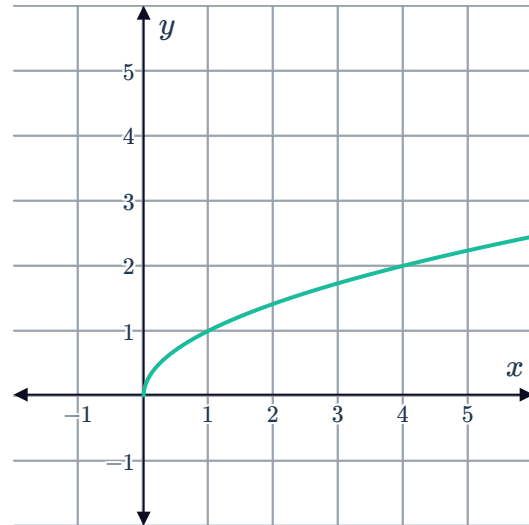
- a Is $f(x)$ continuous over its domain $(-\infty, \infty)$?
- b State the largest interval over which $f(x)$ is continuous.





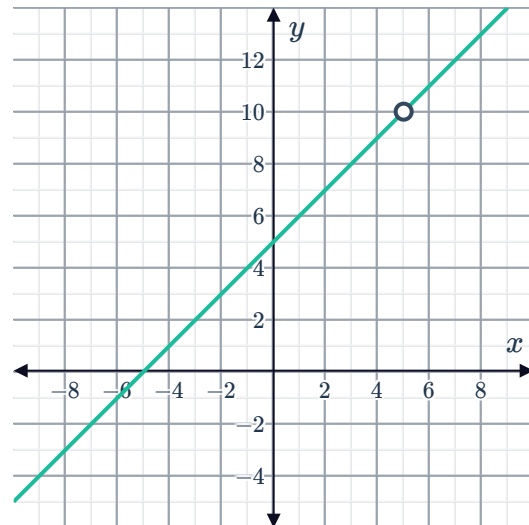
20 Consider the graph of the function $f(x) = \sqrt{x}$:

- a** State the domain of $f(x)$.
- b** State the largest interval over which $f(x)$ is continuous.
- c** Is $f(x) = \sqrt{x}$ a continuous function over its domain?



21 Consider the graph of the function $f(x) = \frac{x^2 - 25}{x - 5}$:

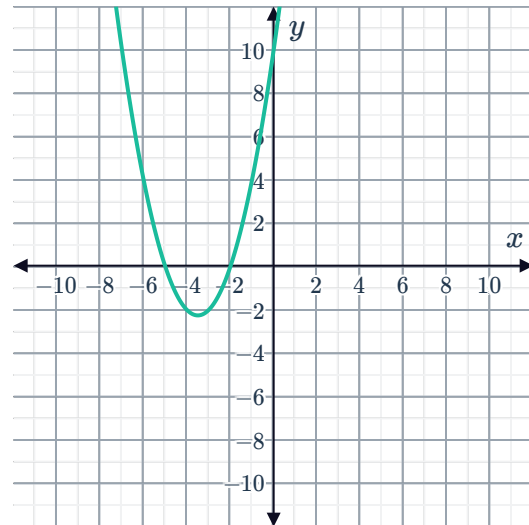
- a** State the domain of $f(x)$.
- b** State the largest interval over which $f(x)$ is continuous.
- c** Is $f(x) = \frac{x^2 - 25}{x - 5}$ a continuous function over its domain?



22 Consider the graph of the function $f(x) = x^2 + 7x + 10$:

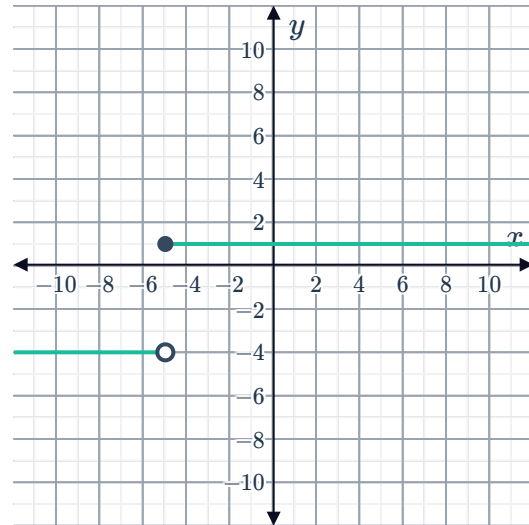


- a State the domain of $f(x)$.
- b State the largest interval over which $f(x)$ is continuous.
- c Is $f(x) = x^2 + 7x + 10$ a continuous function over its domain?



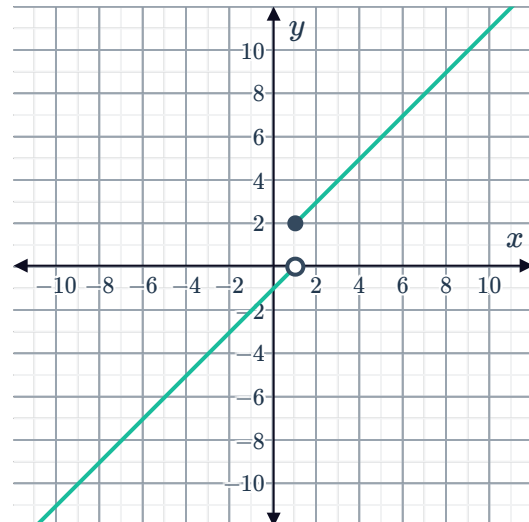
23 Consider the step function $f(x)$ shown:

- a State the domain of $f(x)$.
- b State the largest interval over which $f(x)$ is continuous.
- c Is $f(x)$ a continuous function over its domain? Explain your answer.



24 Consider the function $f(x)$ shown:

- a State the domain of $f(x)$.
- b State the largest interval over which $f(x)$ is continuous.
- c Is $f(x)$ a continuous function over its domain? Explain your answer.





Applications

25 The function $f(n)$ is used to determine the area of a square given its side length.

a State whether the following values are part of the domain of the function:

i -8

ii 6.5

iii $9\frac{1}{3}$

iv $\sqrt{78}$

b For $n \geq 0$, state the area function for a side length of n .

c Sketch the graph of the function f .

26 The number of bees in a colony is initially measured and left to form a hive. The number of bees in the hive is measured each day over the course of one week. The function

$$H(x) = 30(3)^x$$

is found to model the number of bees, H , after x days.

a What is the domain of the function?

b What is the initial number of bees in the colony?

c Can the function be used to model the number of bees over an entire season?

27 The water level in a dam is measured to be 190 yd. After two weeks, the water level is measured again and found to be 140 yd.

a Is there a time during the two weeks that the water level of the dam is at 160 yd?

b To answer part (a) what property do we assume about the water level over the two weeks?

c Consider a function $f(x)$ with domain $[0, 2]$, where $f(0) = 190$ and $f(2) = 140$. If the function is known to be discontinuous, does the function take a value of 160 at some point in its domain? Explain your answer.